Project

Dark Web Forensics – Artifact Analysis and Anti-Forensics Investigation

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Submission Date: 30 June 2025

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1. Introduction

This assignment focuses on exploring dark web activities through Tor Browser and performing forensic analysis on captured memory artifacts. The objective is to detect signs of anti-forensic activities and analyze digital traces left behind by obfuscation methods.

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2. Tools Used & Justification

Tool	Purpose
Tor Browser	Anonymous browsing of dark web websites
FTK Imager / Belkasoft	RAM acquisition from live system
WinPrefetchView	Analysis of `.pf` files to detect Tor activity
Netstat (CMD)	Identify live connections possibly linked to Tor sessions
Bulk Extractor	Extract hidden/cached PII and evidence from RAM
(Optional) Volatility	Deeper memory structure analysis (processes, connections, plugins, etc)

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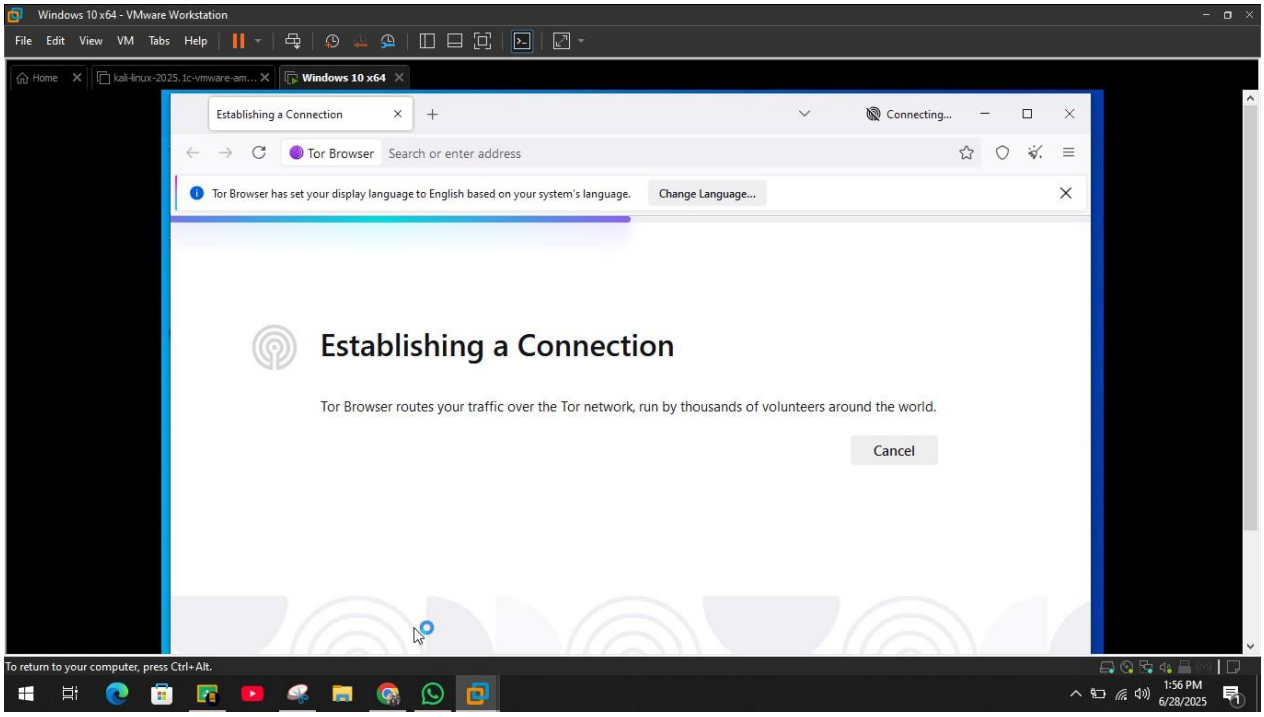
3. Tor Browser Setup & Observations

- Tor Browser was installed on a Windows VM (1 GB RAM).
- Visited the following dark web directories and search engines:
  - Ahmia (https://ahmia.fi)
  - Hidden Wiki
  - Torch
  - Excavator
- Searched for **"Hacker for Hire"** and browsed multiple illicit-themed onion websites.
- ⚠□ **No personal data was shared; all interactions were observational only.**


Windows 10 x64 - VMware Workstation
File Edit View VM Tabs Help
Home
kali-linux-2025.1c-vmware-am...
Windows 10 x64
Establishing a Connection
Connecting...
Tor Browser
Search or enter address
Tor Browser has set your display language to English based on your system's language.
Change Language...
Establishing a Connection
Tor Browser routes your traffic over the Tor network, run by thousands of volunteers around the world.
Cancel
To return to your computer, press Ctrl+Alt.
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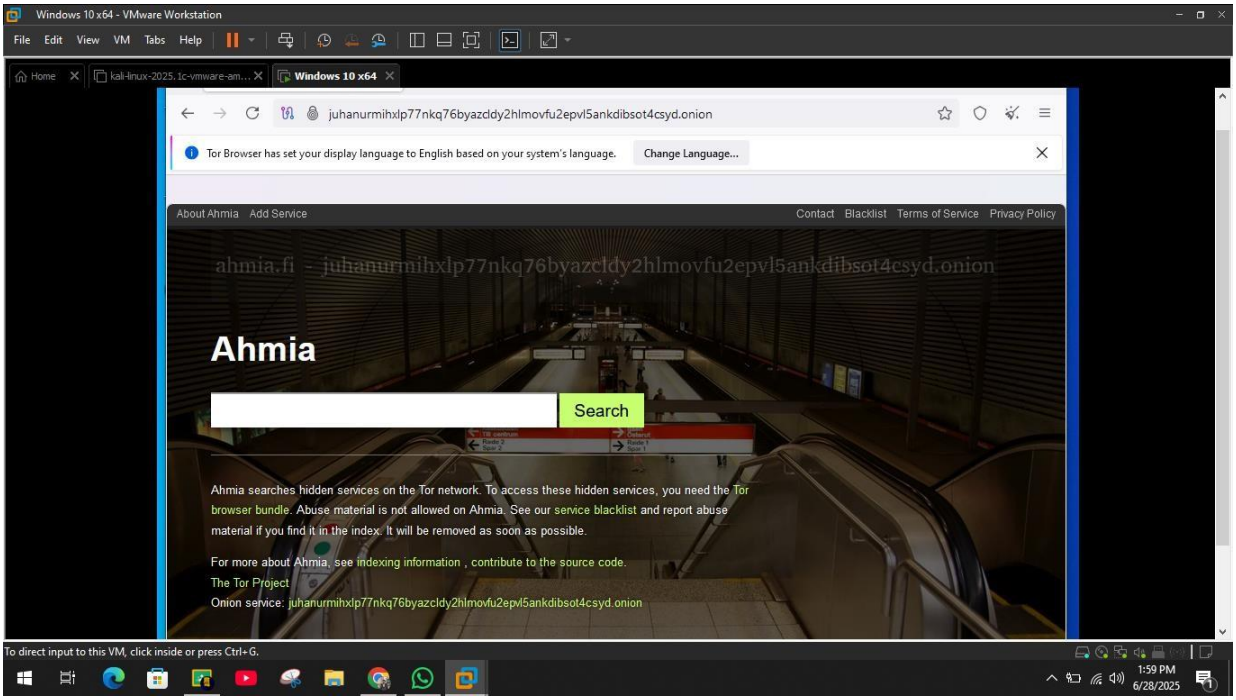
6/28/2025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Windows 10 x64
juhanurmihxlp77nkq76byazcldy2hlmovfu2epvl5ankdibsot4csyd.onion
Tor Browser has set your display language to English based on your system's language.
Change Language...
About Ahmia
Add Service
Contact
Blacklist
Terms of Service
Privacy Policy
ahmia.fi - juhanurmihxlp77nkq76byazcldy2hlmovfu2epvl5ankdibsot4csyd.onion
Ahmia
Search
Ahmia searches hidden services on the Tor network. To access these hidden services, you need the Tor browser bundle. Abuse material is not allowed on Ahmia. See our service blacklist and report abuse material if you find it in the index. It will be removed as soon as possible.
For more about Ahmia, see indexing information , contribute to the source code.
The Tor Project
Onion service: juhanurmihxlp77nkq76byazcldy2hlmovfu2epvl5ankdibsot4csyd.onion
To direct input to this VM, click inside or press Ctrl+G.
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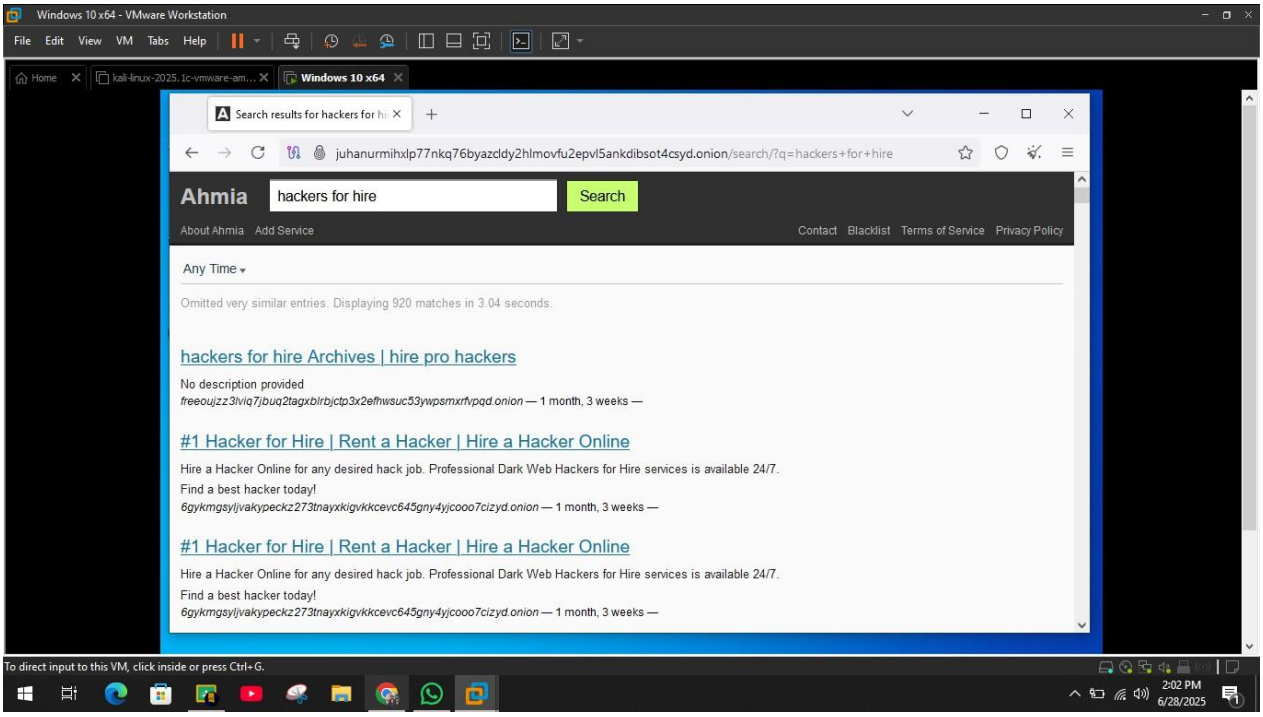
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Search results for hackers for hi
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Ahmia
hackers for hire
Search
About Ahmia
Add Service
Contact
Blacklist
Terms of Service
Privacy Policy
Any Time
Omitted very similar entries. Displaying 920 matches in 3.04 seconds.
hackers for hire Archives | hire pro hackers
No description provided
freeoujzz3lviq7jbuq2tagxblrbjctp3x2efhwsuc53ywpsmxrfvpqd.onion — 1 month, 3 weeks —
#1 Hacker for Hire | Rent a Hacker | Hire a Hacker Online
Hire a Hacker Online for any desired hack job. Professional Dark Web Hackers for Hire services is available 24/7.
Find a best hacker today!
6gykmgsyljvakypeckz273tnayxkigvkkcevc645gny4yjcooo7cizyd.onion — 1 month, 3 weeks —
#1 Hacker for Hire | Rent a Hacker | Hire a Hacker Online
Hire a Hacker Online for any desired hack job. Professional Dark Web Hackers for Hire services is available 24/7.
Find a best hacker today!
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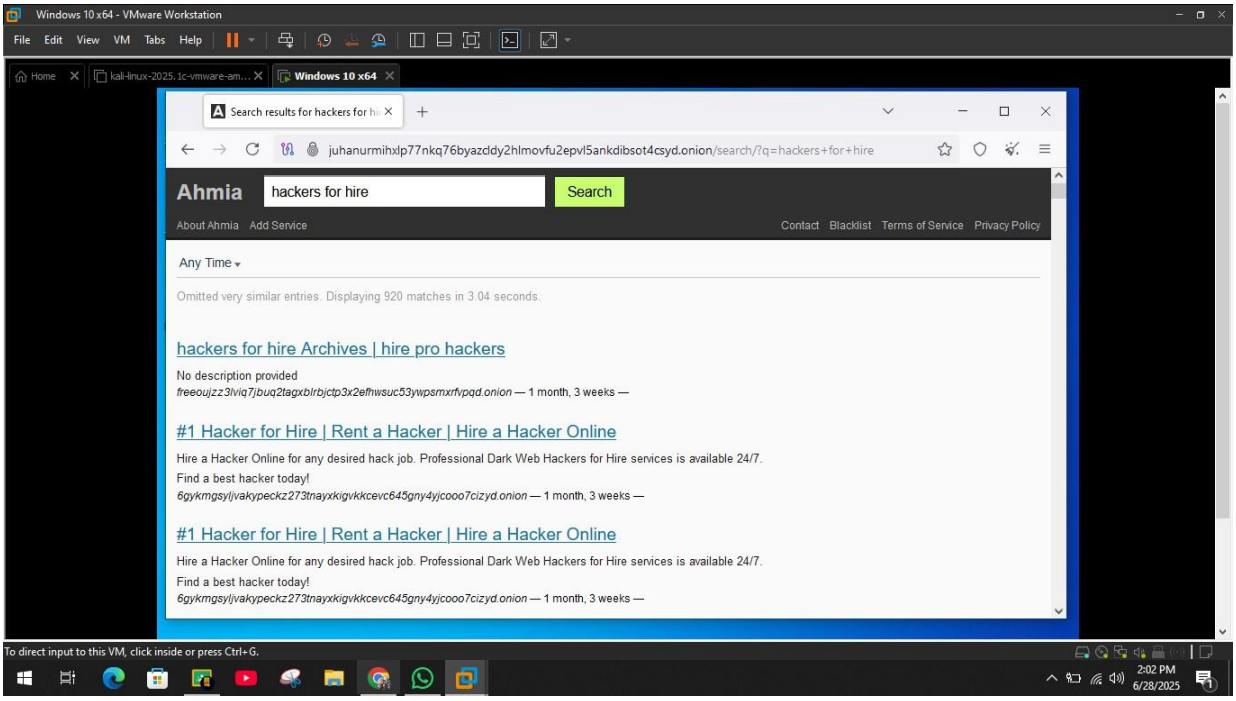
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Search results for hackers for hi
juhanurmihxlp77nkq76byazcldy2hlmovfu2epvl5ankdibsot4csyd.onion/search/?q=hackers+for+hire
Ahmia
hackers for hire
Search
About Ahmia
Add Service
Contact
Blacklist
Terms of Service
Privacy Policy
Any Time
Omitted very similar entries. Displaying 920 matches in 3.04 seconds.
hackers for hire Archives | hire pro hackers
No description provided
freeoujzz3lviq7jbuq2tagxblrbjctp3x2efhwsuc53ywpsmxrfvpqd.onion — 1 month, 3 weeks —
#1 Hacker for Hire | Rent a Hacker | Hire a Hacker Online
Hire a Hacker Online for any desired hack job. Professional Dark Web Hackers for Hire services is available 24/7.
Find a best hacker today!
6gykmgsyljvakypeckz273tnayxkigvkkcevc645gny4yjcooo7cizyd.onion — 1 month, 3 weeks —
#1 Hacker for Hire | Rent a Hacker | Hire a Hacker Online
Hire a Hacker Online for any desired hack job. Professional Dark Web Hackers for Hire services is available 24/7.
Find a best hacker today!
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To direct input to this VM, click inside or press Ctrl+G.
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6/28/2025

Windows 10 x64 - VMware Workstation

File Edit View VM Tabs Help

Home | kali-linux-2025.1c-vmware-am... | Windows 10 x64

The Hidden Wiki

https://thehidden-wiki.org/wiki/index.php/Main_Page

Log in

The Hidden Wiki

Main page | Discussion | Read | View source | Search The Hidden Wiki

Main Page

Main page
Random page
Help about MediaWiki

Tools

What links here
Special pages
Printable version
Permanent link

Welcome to The Hidden Wiki! Our official Hidden Wiki url in 2024 is: http://

Contents [hide]

1 Editor's picks
2 Volunteer
3 Introduction Points
4 Financial Services
5 Commercial Services
6 Domain Services
7 Anonymity & Security
8 Darknet versions of popular sites
9 Blogs / Essays / News Sites
10 General Knowledge
11 Email / Messaging
12 Social Networks
13 Forums / Boards / Chats
14 Whistleblowing

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Windows 10 x64 - VMware Workstation

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The Dark Web Pug

jgwe5qqdbyvudjqskaajbfibfewew4pndx52dye7ug3mt3jimmktkid.onion

More wow.

http://prjd5pmbug2cnfs67s3y65ods27vamswdaw2lnwf45ys3pjl55h2gwqd.onion/ Dark Web Hackers for hire
http://55niksbd22qqaedkw36qw4cpofmbxdtbwonxam7ov2ga62zqbhgty3yd.onion/ AccMarket - Premium Paypal, Ebay and bank accounts
http://s57divisqlcjtsyutxjz2ww77vlbwpxgodtijcsrgsuts4js5hnxkhqd.onion/ Cardshop - USA CVV KNOWN BALANCE & Worldwide CC & CVV
http://jbtb75gqlr57qurikzy2bxxjftzkmanynesmoxbzzcp7qf5t46u7ekqd.onion/ Darkmining - Bitcoin mining with stolen electricity
http://jhi4v5rjly75ggha26cu2eeyfhwvgbde4w6d75vepwxt2zht5sqfhuqd.onion/ Bitcoin Investment Trust - earn 5-9% per week!
http://rxmyl3izgquew65nicavsk6loyyblztng6puq42firpvbe32sefvnbad.onion/ Mobile Store - Best unlocked cell phones vendor
http://vhlehwexxmbnvecbmsk4ormttdvhlhbnyabai4cithvizzaduf3gmayd.onion/ Kamagra 4 Bitcoin - Like Viagra but cheaper

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Windows 10 x64 - VMware Workstation

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Dark Web Hackers

prjd5pmbug2cnfs67s3y65ods27vamswdaw2lnwf45ys3pjl55h2gwqd.onion

Dark Web Hackers

Products Register Login

Welcome to the Dark Web Hackers

Have you tried to buy hacking services on the dark web before? Not happy with the results? Only empty promises but no one getting the job done?

Then you should try Vladimir and George, the dark webs most trusted hackers for getting things done.

Unlike others, our prices are not the cheapest, but if we can't do a job, you will get a full refund!

Vladimir

Hello, my name is Vladimir.
I am the technical expert at dark web hackers.

My expertise is programming, running exploits, setting up DDOS attacks and i like the challenge of doing things where most others give up.
I can "recover" passwords of most social networks easily, remote control smartphones, and most other things that are useful because i spent years to find methods that really work.
Here you can find a list of my services, if it is not listed, then minimum price will be

To direct input to this VM, click inside or press Ctrl+G.

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Windows 10 x64 - VMware Workstation

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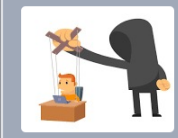
Dark Web Hackers

prjd5pmbug2cnfs67s3y65ods27vamswdaw2lnwf45ys3pjl55h2gwqd.onion

Buy now

George



Hello, my name is George.
My hacking skills are not as perfect as Vladimir's, but i am really good with social engineering.
And i really like messing with people, i don't care what you want to do to them.
If there is something i can't do then Vladimir will help and teach me for next time.

Product	Price	Quantity
Destroying someones life: Your target will have legal problems or financial problems, proven methods including child porn that always works	1700 USD = 0.01574 ฿	1 X Buy now
Spreading false information about someone on social media, not as life ruining but still nasty	450 USD = 0.00417 ฿	1 X Buy now

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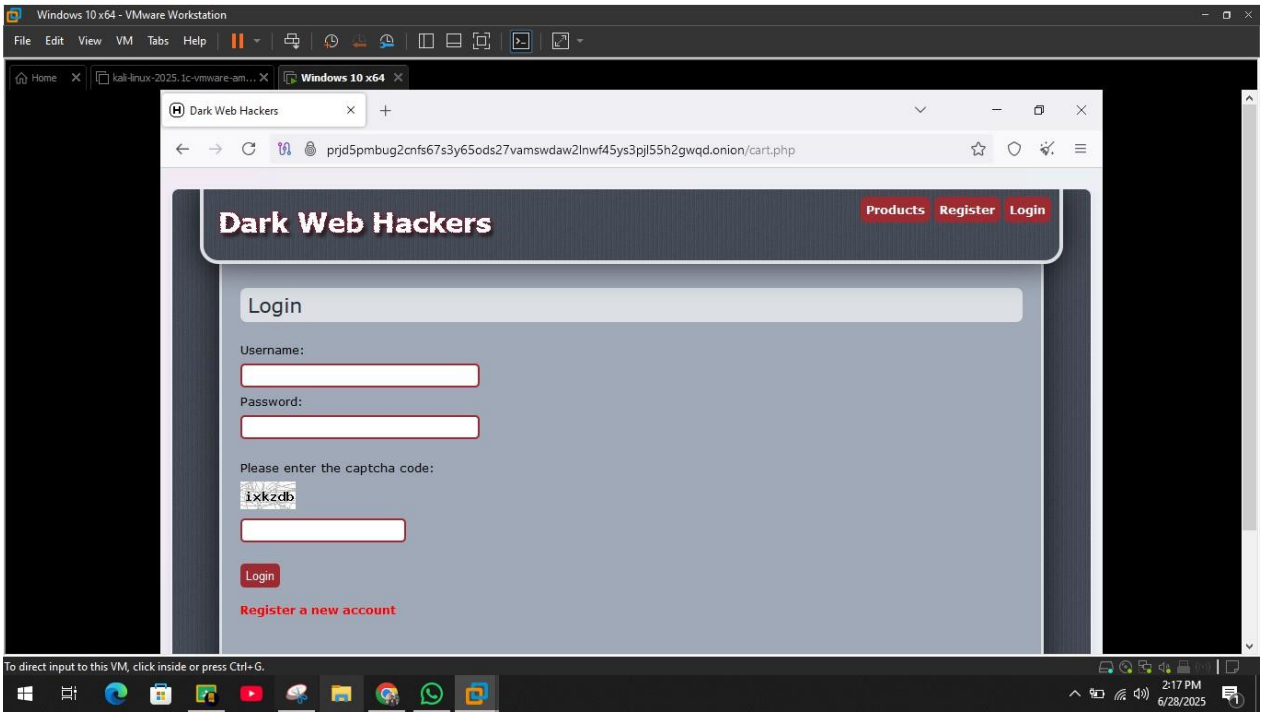
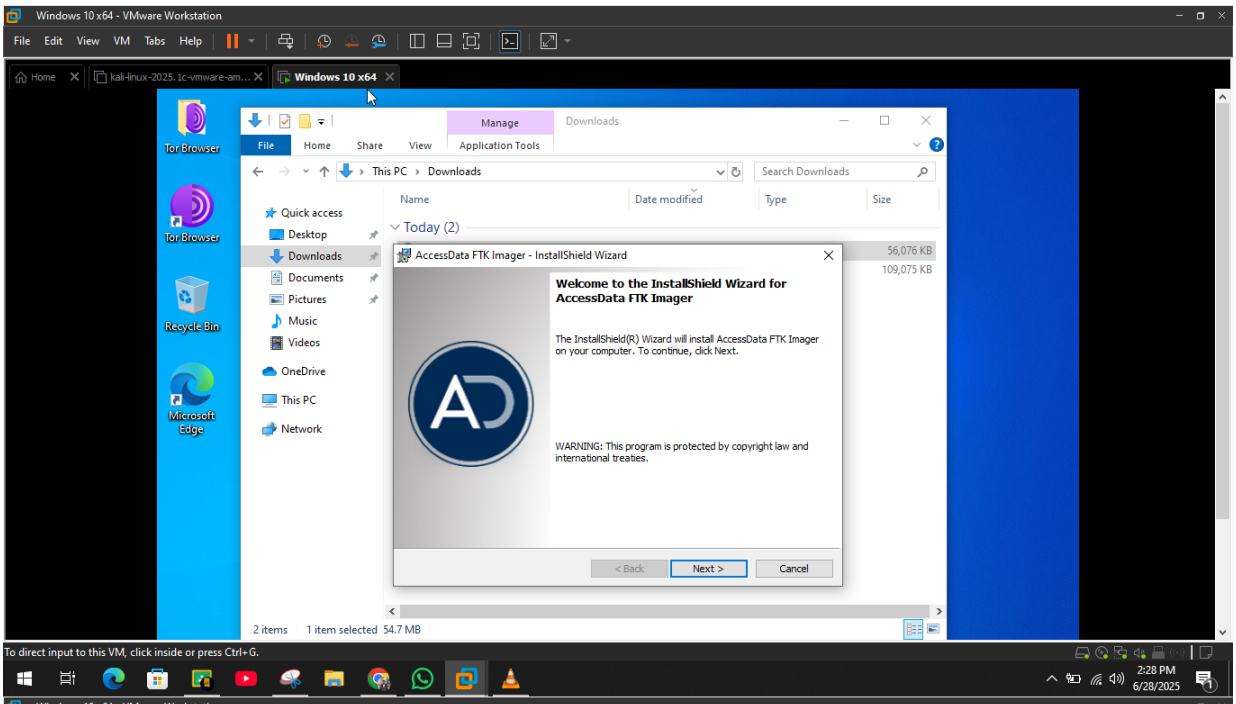


4. RAM Acquisition Process

- After closing the Tor browser, FTK Imager was used to acquire a full memory dump.
- Output file saved as `tor_session.mem`.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Home
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Windows 10 x64
Tor Browser
Recycle Bin
Microsoft Edge
Manage
Downloads
File Home Share View Application Tools
This PC › Downloads
Search Downloads
Quick access
Desktop
Downloads
Documents
Pictures
Music
Videos
OneDrive
This PC
Network
Name
Date modified
Type
Size
Today (2)
56,076 KB
109,075 KB
AccessData FTK Imager - InstallShield Wizard
Welcome to the InstallShield Wizard for AccessData FTK Imager
The InstallShield(R) Wizard will install AccessData FTK Imager on your computer. To continue, click Next.
WARNING: This program is protected by copyright law and international treaties.
< Back
Next >
Cancel
2 items
1 item selected 54.7 MB
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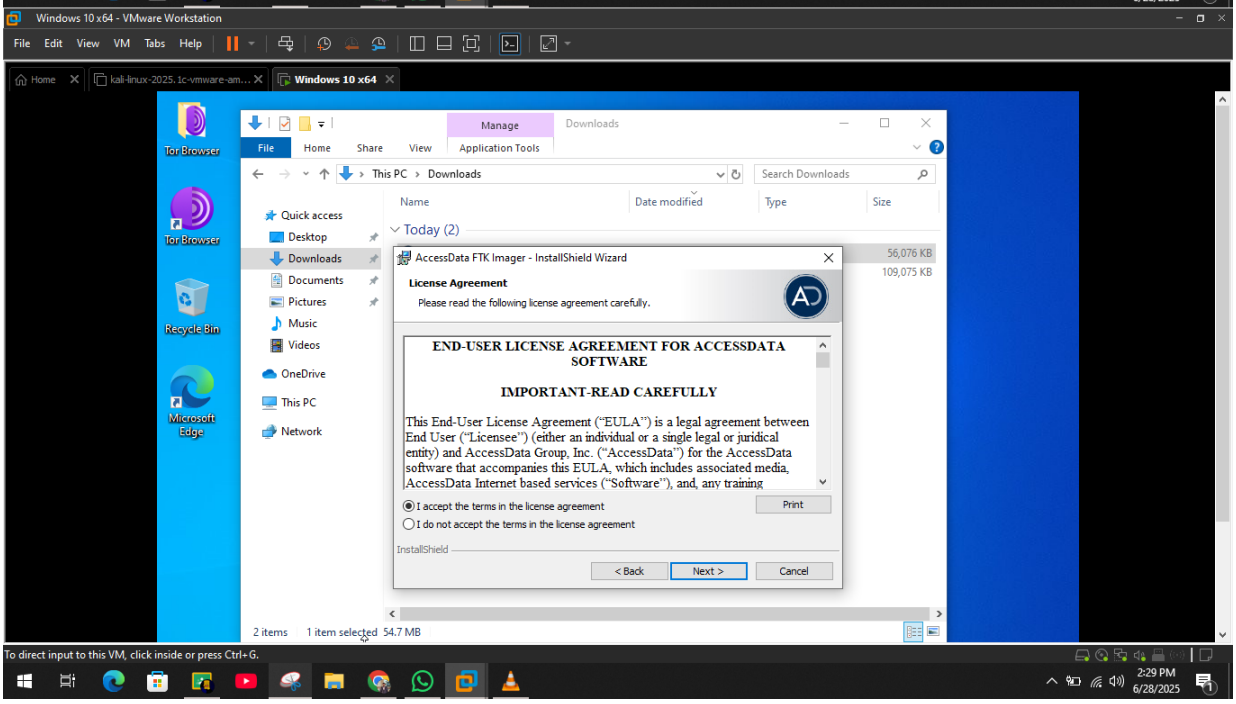
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Home
kali-linux-2025.1c-vmware-am...
Windows 10 x64
Tor Browser
Recycle Bin
Microsoft Edge
Manage
Downloads
File Home Share View Application Tools
This PC › Downloads
Search Downloads
Quick access
Desktop
Downloads
Documents
Pictures
Music
Videos
OneDrive
This PC
Network
Name
Date modified
Type
Size
Today (2)
56,076 KB
109,075 KB
AccessData FTK Imager - InstallShield Wizard
License Agreement
Please read the following license agreement carefully.
END-USER LICENSE AGREEMENT FOR ACCESSDATA SOFTWARE
IMPORTANT-READ CAREFULLY
This End-User License Agreement ("EULA") is a legal agreement between End User ("Licensee") (either an individual or a single legal or juridical entity) and AccessData Group, Inc. ("AccessData") for the AccessData software that accompanies this EULA, which includes associated media, AccessData Internet based services ("Software"), and, any training
I accept the terms in the license agreement
I do not accept the terms in the license agreement
Print
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2 items
1 item selected 54.7 MB
To direct input to this VM, click inside or press Ctrl+G.
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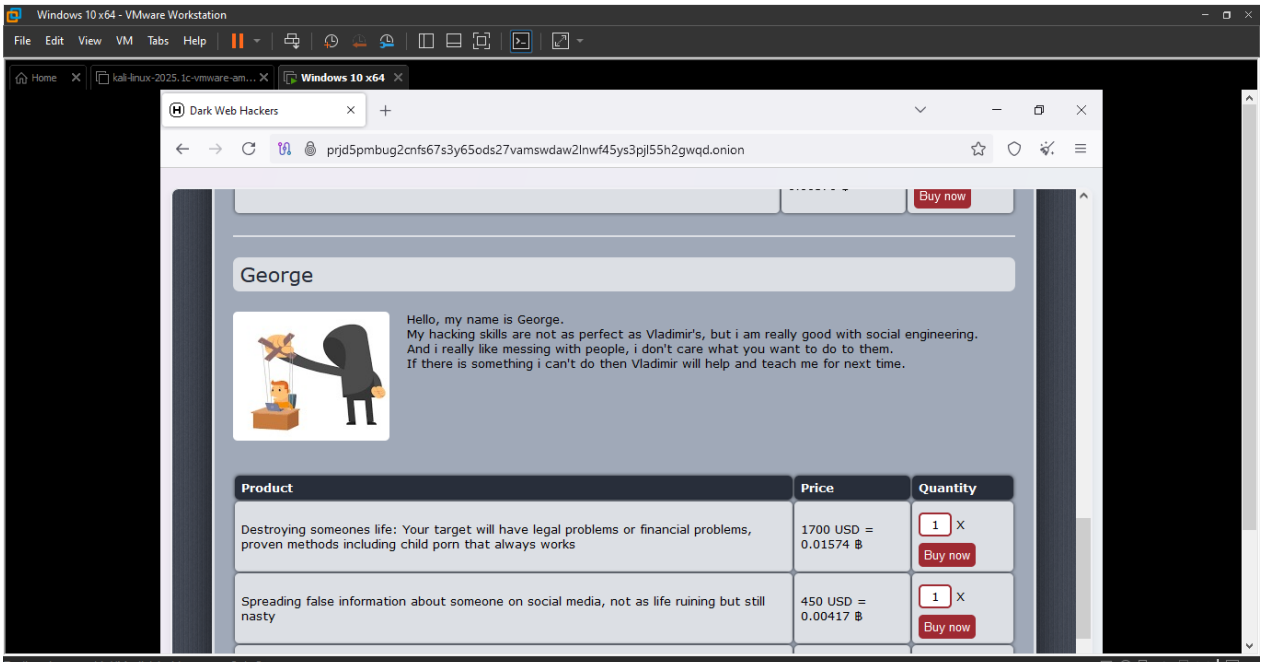
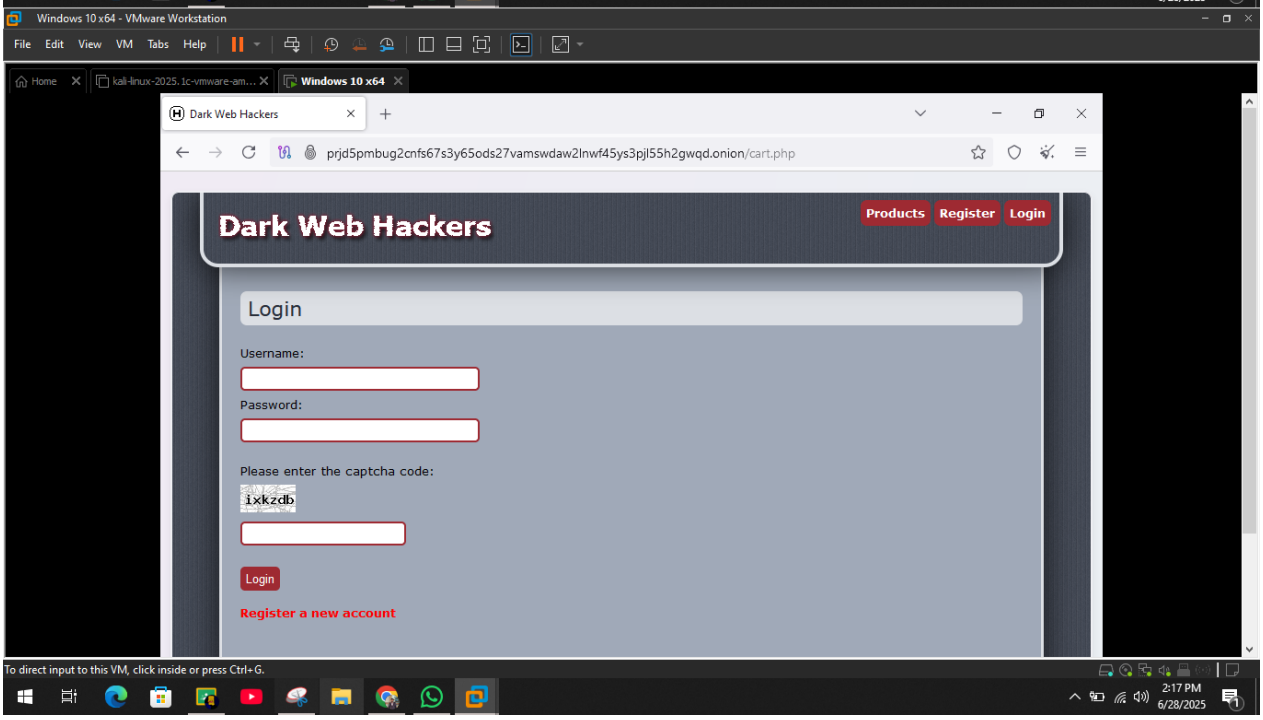
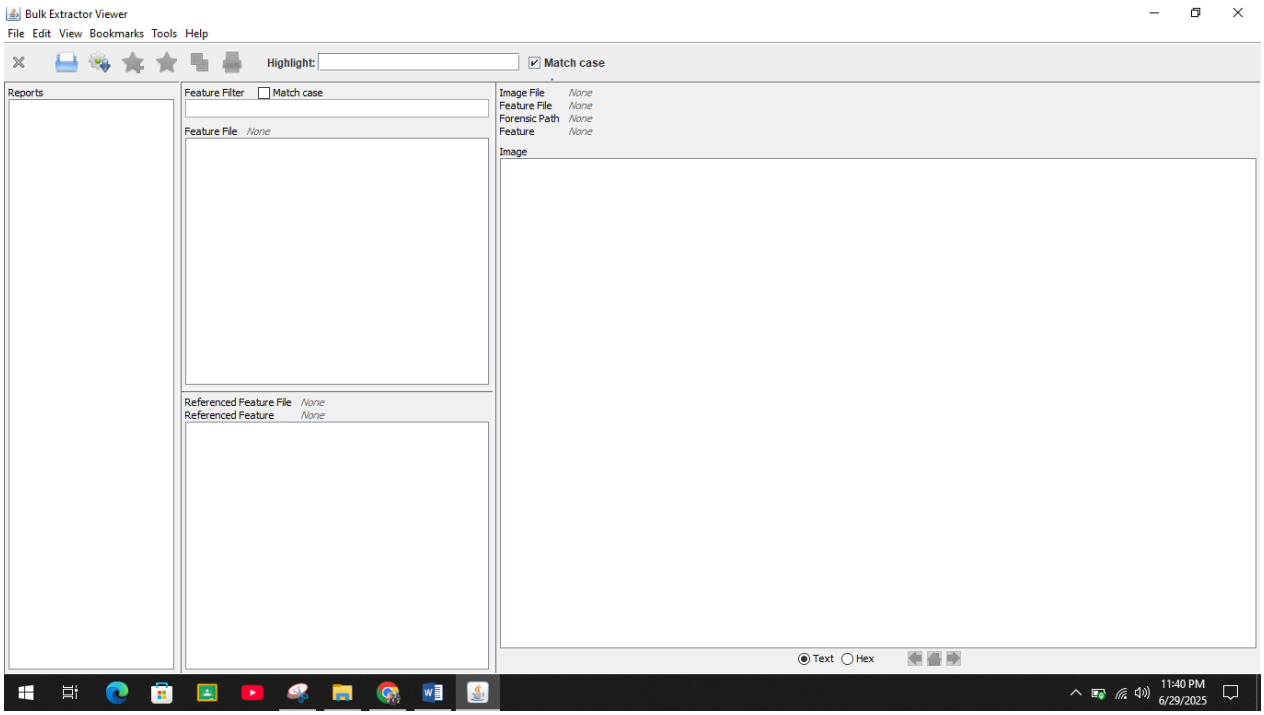
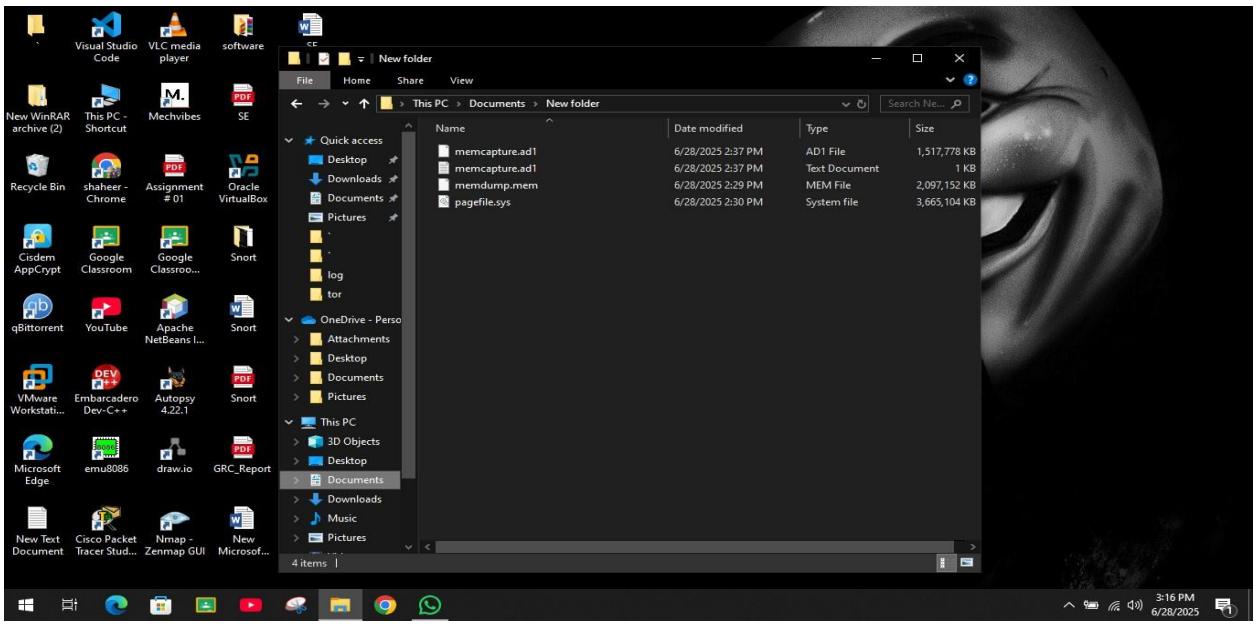




6. Memory Analysis Using Bulk Extractor

- RAM dump processed with Bulk Extractor.
- Extracted data:
  - `.onion` domains
  - Email addresses
  - Strings resembling credit card numbers
- **Optional Volatility** use revealed hidden processes possibly related to obfuscation.





7. Detection of Anti-Forensic Behavior

- Signs of anti-forensics identified:
  - No browser history present
  - Some memory patterns showed zeroed-out data
  - Volatility indicated a hidden/terminated process linked to Tor
- Indicates user-level or software-level obfuscation attempts

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8. Timeline of Events

Time	Event Description
4:00 PM	VM Booted, Tor Browser Installed
4:30 PM	Browsing activity on dark web
5:00 PM	Tor closed, RAM acquired
5:15 PM	Prefetch and netstat analysis completed
6:00 PM	RAM analysis via Bulk Extractor and tools

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9. Challenges and Mitigation Strategies

Challenge	Solution
Slow VM performance	Increased allocated RAM to 1.5 GB
Tor not loading some onion sites	Switched to alternate onion directories
FTK Imager freezing during dump	Used Belkasoft RAM Capturer as backup
Misleading artifacts from normal apps	Filtered based on time and process names

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10. Conclusion

This assignment helped solidify practical understanding of dark web browsing behavior, memory acquisition, and forensic evidence extraction. It also revealed how attackers use anti-forensic techniques like memory wiping and browser history obfuscation to hide activity.

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11. References

- Tor Project: https://www.torproject.org/
- FTK Imager: https://accessdata.com
- Bulk Extractor: https://digitalcorpora.org
- Volatility Framework: https://www.volatilityfoundation.org
- WinPrefetchView: https://www.nirsoft.net